Q-4) Discuss SAR of alcohols & their derivatives

ALCOHOLS

sedative hypnotics (miscellaneous)

Actions: Acts at the GABAA receptor to potentiate the action of GABA.
Also inhibits NMDA (N-Methyl-D-Asparate) glutamate receptors.
These actions result in an array of actions including CNS depression.

SAR for hypnotic activity of alcohols

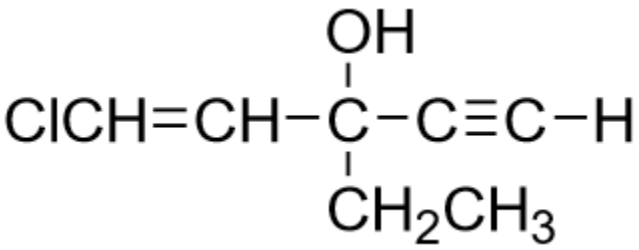
- Activity increases as the chain length increase with maximum at C8
- Branching the chain increases activity

Tertiary > Secondary > Primary

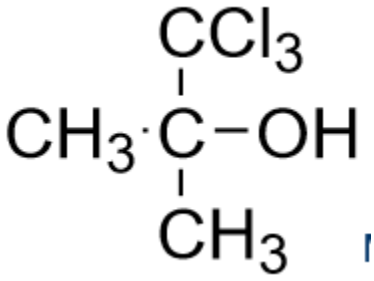
decrease metabolism by oxidation to the corresponding carboxylic acids

- **Replacing hydrogen with a halogen increases activity**

Ethchlorvynol and Chlorobutanol
– A tertiary, lipophilic alcohol



Ethchlorvynol (1-chloro-3-ethyl-1-penten-4-yn-3-ol)



Chlorobutanol

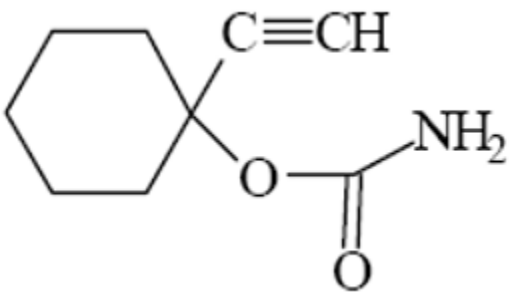
Alcohol derivatives

- ✓ **Carbamates**

Carbamylation increases depressant potency

- **Carbamates-** much more resistant to metabolic inactivation than hydroxyl functions

The best example is Ethinamate which is Carbamate analogue of acetylenic alcohol.



- It is a short-acting hypnotic to which tolerance rapidly develops; it is indicated only for periods of 7 days or less.